

Surprise Quiz 07

Q1. What is correct about POSIX pipes?

- A. Two way communication channel between processes
- B. It creates child process and creates a communication channel with parent
- C. It behaves similar to a garden hose pipe connected from a tap to the plants (correct)**
- D. It is a system call (correct)**

Q2. What is correct about signals?

- A. Unlimited form of IPC
- B. Requires help from OS for every signalling instances (correct)**
- C. Hardware interrupts
- D. Communication between two processes inside userspace

Q3. Which operations are part of the POSIX shared memory lifecycle?

- A. shm_open() creates/opens the memory object (correct)**
- B. mmap() maps the shared object into process address space (correct)**
- C. ftruncate() sets the byte size of the segment (correct)**
- D. pipe() initializes the shared memory buffer

Q4. What is true about POSIX shared Memory?

- A. Two-way communication channel without system call overhead during data transfer (correct)**
- B. Kernel automatically synchronizes concurrent reads and writes
- C. Creates a child process and maps its address space to the parent
- D. Managed entirely inside user space for every read/write operation (correct)**

Q5. Which of the following is correct regarding file descriptors in the following:

```
int pipe(int pipefd[2]);
```

- A. pipefd[0] is for reading, pipefd[1] is for writing (correct)**
- B. pipefd[0] is for writing, pipefd[1] is for reading
- C. Both descriptors permit simultaneous reading and writing
- D. Reader/Writer process must close one of the index in pipefd (correct)**

Q6. Which operations are associated with signal management in C?

- A. kill() is used to send a signal to a process (correct)**
- B. sigaction() is used to register a signal handler function (correct)**
- C. signal.h defines signal constant macros like SIGINT and SIGCHLD (correct)**
- D. printf() and malloc() can be called safely inside any signal handler